Linear Regression Machine Learning Project for House Price Prediction

Import Libraries

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

Importing Data and Checking out.

```
HouseDF = pd.read_csv('USA_Housing.csv')
```

```
HouseDF.head()
```

```
   Avg. Area Income  Avg. Area House Age  Avg. Area Number of Rooms  \
0      79545.458574             5.682861                   7.009188   
1      79248.642455             6.002900                   6.730821   
2      61287.067179             5.865890                   8.512727   
3      63345.240046             7.188236                   5.586729   
4      59982.197226             5.040555                   7.839388   

   Avg. Area Number of Bedrooms  Area Population         Price  \
0                          4.09     23086.800503  1.059034e+06   
1                          3.09     40173.072174  1.505891e+06   
2                          5.13     36882.159400  1.058988e+06   
3                          3.26     34310.242831  1.260617e+06   
4                          4.23     26354.109472  6.309435e+05   

                                             Address  
0  208 Michael Ferry Apt. 674\nLaurabury, NE 3701...  
1  188 Johnson Views Suite 079\nLake Kathleen, CA...  
2  9127 Elizabeth Stravenue\nDanieltown, WI 06482...  
3                          USS Barnett\nFPO AP 44820  
4                         USNS Raymond\nFPO AE 09386  
```

```
HouseDF.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   Avg. Area Income              5000 non-null   float64
```

```
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

```
HouseDF.describe()
```

```
       Avg. Area Income  Avg. Area House Age  Avg. Area Number of
Rooms  \
count       5000.000000          5000.000000
5000.000000
mean       68583.108984             5.977222
6.987792
std        10657.991214             0.991456
1.005833
min        17796.631190             2.644304
3.236194
25%        61480.562388             5.322283
6.299250
50%        68804.286404             5.970429
7.002902
75%        75783.338666             6.650808
7.665871
max       107701.748378             9.519088
10.759588

       Avg. Area Number of Bedrooms  Area Population         Price
count                   5000.000000      5000.000000  5.000000e+03
mean                       3.981330     36163.516039  1.232073e+06
std                        1.234137      9925.650114  3.531176e+05
min                        2.000000       172.610686  1.593866e+04
25%                        3.140000     29403.928702  9.975771e+05
50%                        4.050000     36199.406689  1.232669e+06
75%                        4.490000     42861.290769  1.471210e+06
max                        6.500000     69621.713378  2.469066e+06
```

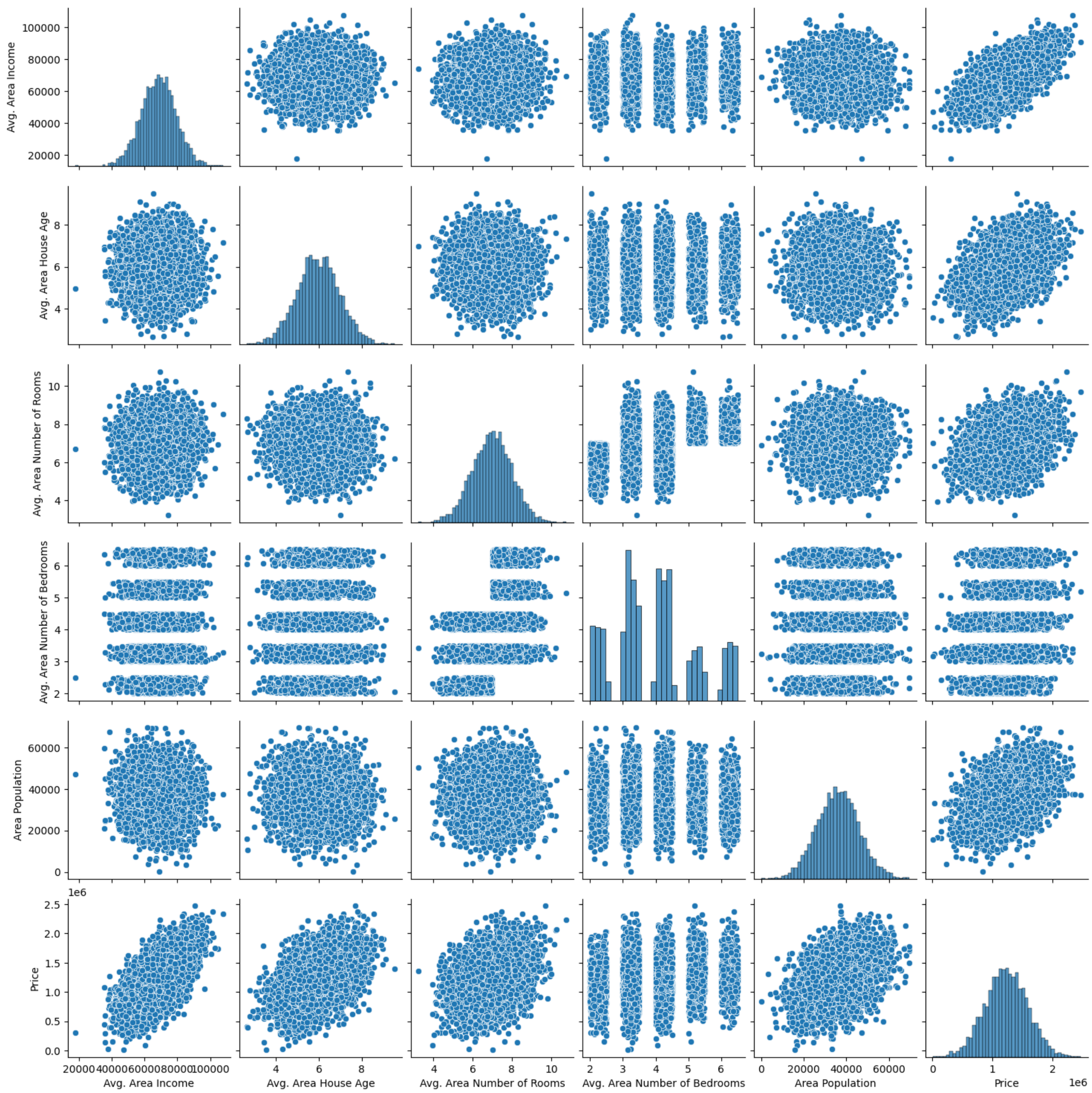
```
HouseDF.columns
```

```
Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of
Rooms',
       'Avg. Area Number of Bedrooms', 'Area Population', 'Price',
'Address'],
      dtype='object')
```

Exploratory Data Analysis for House Price Prediction

```
sns.pairplot(HouseDF)
```

```
<seaborn.axisgrid.PairGrid at 0x21ab401e490>
```




```
sns.distplot(HouseDF['Price'])
```

```
C:\Users\Sam\AppData\Local\Temp\ipykernel_11812\4158129596.py:1:
UserWarning:

`distplot` is a deprecated function and will be removed in seaborn
```

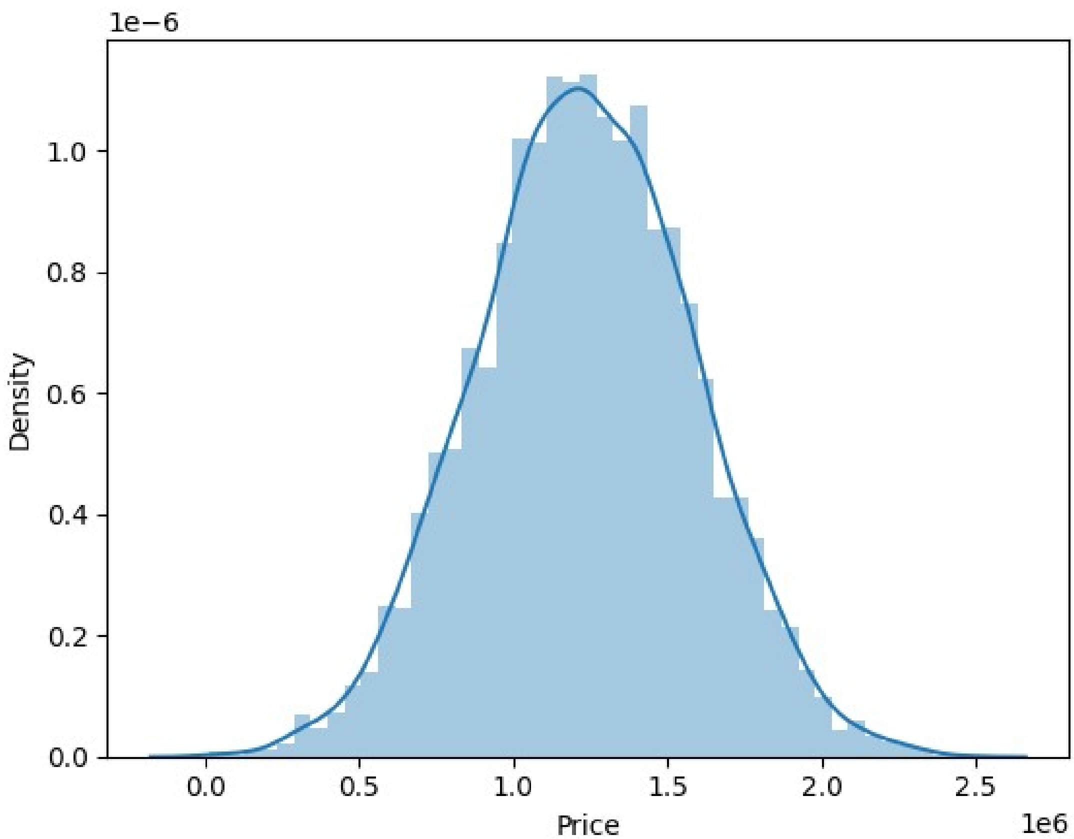
```
v0.14.0.

Please adapt your code to use either `displot` (a figure-level
function with
similar flexibility) or `histplot` (an axes-level function for
histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(HouseDF['Price'])

<Axes: xlabel='Price', ylabel='Density'>
```




```
sns.heatmap(HouseDF.corr(numeric_only=True), annot=True)

<Axes: >
```

Training a Linear Regression Model

X and y List

```
X = HouseDF[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area
Number of Rooms',
               'Avg. Area Number of Bedrooms', 'Area Population']]

y = HouseDF['Price']
```

Split Data into Train, Test

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.4, random_state=101)
```

Creating and Training the LinearRegression Model

```
from sklearn.linear_model import LinearRegression

lm = LinearRegression()

lm.fit(X_train,y_train)

LinearRegression()
```

LinearRegression Model Evaluation

```
print(lm.intercept_)

-2640159.7968526953

coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])
coeff_df
```

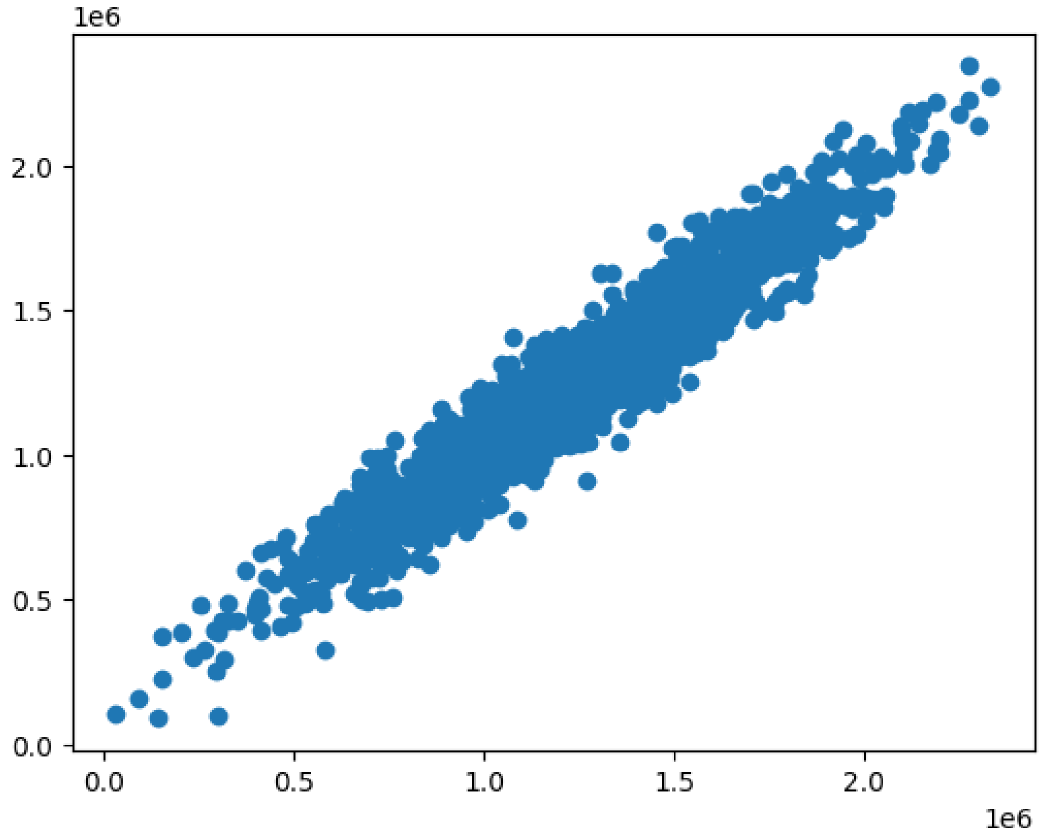
	Coefficient
Avg. Area Income	21.528276
Avg. Area House Age	164883.282027
Avg. Area Number of Rooms	122368.678027
Avg. Area Number of Bedrooms	2233.801864
Area Population	15.150420

Predictions from our Linear Regression Model

```
predictions = lm.predict(X_test)

plt.scatter(y_test,predictions)

<matplotlib.collections.PathCollection at 0x21ab6d49950>
```



In the above scatter plot, we see data is in line shape, which means our model has done good predictions.

```
sns.distplot((y_test-predictions),bins=50);

C:\Users\Sam\AppData\Local\Temp\ipykernel_11812\1326397652.py:1:
UserWarning:

`distplot` is a deprecated function and will be removed in seaborn
v0.14.0.

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  sns.distplot((y_test-predictions),bins=50);
```

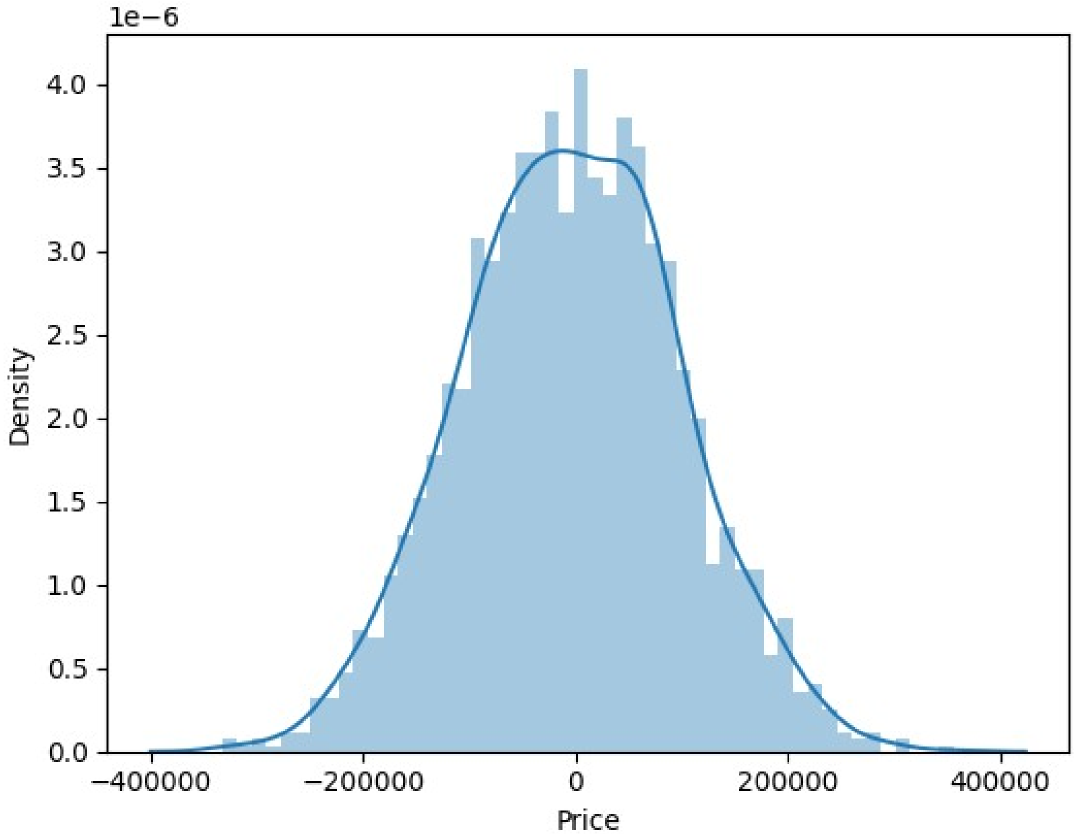

In the above histogram plot, we see data is in bell shape (Normally Distributed), which means our model has done good predictions.

Regression Evaluation Metrics

```
from sklearn import metrics

print('MAE:', metrics.mean_absolute_error(y_test, predictions))
print('MSE:', metrics.mean_squared_error(y_test, predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test,
predictions)))

MAE: 82288.22251914942
MSE: 10460958907.20898
RMSE: 102278.82922290899
```